Summary in Graph

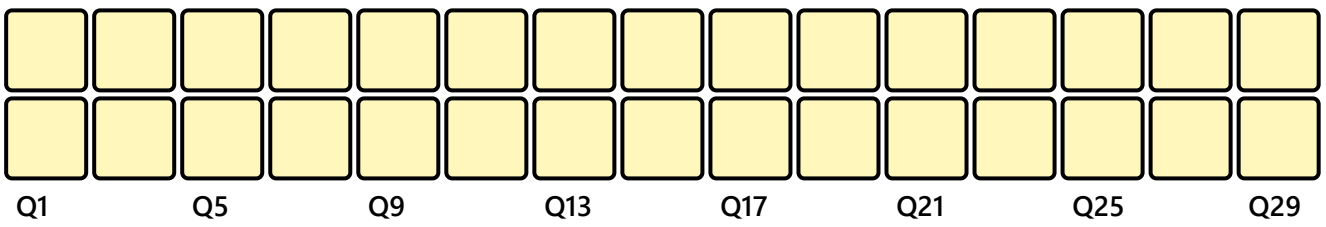
Exam Summary (GATE Overflow CS Test Series 2026 | Discrete Mathematics | Shallow Test 2)

Qs. Attempted:	0 (0 + 0)	Correct Marks:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)	Penalty Marks:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)	Resultant Marks:	0 (0 + 0)

Total Questions:	30 (10 + 20)
Total Marks:	50 (10 + 40)
Exam Duration:	90 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (1) out of 14 users

EXAM RESPONSE EXAM STATS FEEDBACK

Technical

Q #1 Numerical Type Award: 1 Penalty: 0 Discrete Mathematics

The minimum number of socks that must be taken out of a drawer containing 10 pairs of white socks, 8 pairs of black socks, and 5 pairs of blue socks to ensure that at least three pairs of the same color are taken out is ____

Your Answer: | Correct Answer: 16:16 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 Numerical Type | Award: 1 | Penalty: 0 | Discrete Mathematics

Consider a set C with 5 elements. Let P_1 denote the number of injective functions from C to a set D with 6 elements, and let P_2 denote the number of bijective functions from C to C. Calculate $P_1 - P_2$.

Your Answer: | Correct Answer: 600:600 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 Numerical Type | Award: 1 | Penalty: 0 | Discrete Mathematics

How many bit strings of length eight either start with a 0 bit or end with the two bits "11"?.

Your Answer: | Correct Answer: 160:160 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 Multiple Choice Type | Award: 1 | Penalty: 0.33 | Discrete Mathematics

Let $F(z) = \frac{1}{1-3z} = \sum_{k=0}^{\infty} h(k) z^k$ where $|z| < \frac{1}{3}$. $h(k)$ is:

A. 3^k
B. k^3
C. $3^k + k$
D. $\frac{1}{3^k}$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 Numerical Type | Award: 1 | Penalty: 0 | Discrete Mathematics

The number of bit strings of length 5 that do not contain "101" as a substring is ____

Your Answer: | Correct Answer: 21:21 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 Numerical Type | Award: 1 | Penalty: 0 | Discrete Mathematics

The number of positive integers less than 500 that are divisible by 5 but not by 3 is ____

Your Answer: | Correct Answer: 66:66 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 Multiple Choice Type | Award: 1 | Penalty: 0.33 | Mathematical Logic

How many 9-digit numbers(with the available digits in the given number) greater than 123456789 have no two digits the same?

A. $8! + 7!$
B. $9!$
C. $9! - 1$
D. $9! + 2$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Discrete Mathematics

How many bit strings of length 6 contain more 1's than 0's?

A. 20
B. 15
C. 25
D. None of these

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Discrete Mathematics

Twelve employees completed a training program consisting of 12 modules, where each module carries 1 point. The total sum of their points came out to be 80. Now consider the following two statements:
S_1: All the point totals are different.
S_2: There are at least two employees with the same point total.
Which of the above statements is/are TRUE?

A. Only Statement S_1 is true

B. Only Statement S_2 is true

C. Both S_1 and S_2 can be true

D. Both S_1 and S_2 are false

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Mathematical Logic

Let $T1 = \{4, 5, 6, \ldots, 14\}$ and $T2 = \{k, q, r, s, t, u, v, w, x, y, z\}$. The number of onto (surjective) functions from $T2 \rightarrow T1$ when $f(k) = 8$ is __.

A. $11! \left\{ {11 \atop 11} \right\}$

B. $11! \left\{ {11 \atop 10} \right\}$

C. $10! \left\{ {10 \atop 10} \right\}$

D. 1

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 | Numerical Type | Award: 2 | Penalty: 0 | Discrete Mathematics

Professor Nivedita has 7 distinct books which she decided to distribute among her 5 students. She hires a consultancy firm for the same which charges Rs.50 for each of the possible distributions, assuming each student gets at least one book and all students being considered distinct. How much fee will Professor have

to pay the agency (in multiples of 50)?.

Your Answer: | Correct Answer: 16800:16800 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 Numerical Type | Award: 2 | Penalty: 0 | Mathematical Logic

A game of cards is played among 4 friends. After the initial split of 52 cards, the oldest player announced that he has received 3 Kings and all the remaining 10 cards are from the same suit. The total number of ways in which this is possible is ___.

Your Answer: | Correct Answer: 1056:1056 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Discrete Mathematics

For the National Spelling Bee Competition, 25 participants are selected from 5 regions: 4 from East, 5 from West, 6 from North, 5 from South, and 5 from Central, with each region having a representative. From these participants, a final team of 10 will be selected. What is the probability that all the regions will be represented in the final team by their representatives?

A. $\frac{\binom{20}{5}}{\binom{25}{10}}$

B. $\frac{\binom{18}{5}}{\binom{25}{10}}$

C. $\frac{\binom{20}{5}}{\binom{25}{8}}$

D. $\frac{\binom{20}{4}}{\binom{25}{10}}$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Mathematical Logic

Match the sequences on the left with their corresponding generating functions on the right:

Sequences		Generating Functions	
A.	$1, 3, 5, 7, 9, 11, \ldots$	**P.**	$\frac{(1+x)}{(1-x)^2}$
B.	$0, 1, 2, 3, 4, 5, 6, \ldots$	**Q.**	$\frac{x}{(1-x)^2}$
C.	$0, 1, 4, 9, 16, 25, 36, \ldots$	**R.**	$\frac{x(1+x)}{(1-x)^3}$
D.	$1, 1, 2, 6, 24, 120, 720, \ldots$	**S.**	No Ordinary GF that converges

A. A - P, B - Q, C - R, D - S

B. A - P, B - R, C - S, D - Q

C. A - Q, B - R, C - S, D - P

D. None of the above

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 | Numerical Type | Award: 2 | Penalty: 0 | Discrete Mathematics

Find the number of positive integer solutions to the equation $3x + 2y + z = 20$

Your Answer: | Correct Answer: 24:24 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #16 | Numerical Type | Award: 2 | Penalty: 0 | Discrete Mathematics

A rectangular field is divided into a grid of 8 rows and 12 columns of smaller rectangles. The total number of rectangles (of all sizes) that can be formed within this grid is ____.

Your Answer: | Correct Answer: 2808:2808 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #17 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Discrete Mathematics

The generating function of the sequence $1, 4, 9, 16, 25, \ldots$ is:

$$\frac{x(1+x)}{(1-x)^3}$$

. The sequence generated by the generating function $\frac{1}{1-2x-3x^2}$ is _____.

A. $1, 2, 7, 24, 85, \ldots$

B. $1, 2, 5, 10, 20, \ldots$

C. $1, 2, 6, 20, 70, \ldots$

D. $1, 2, 7, 20, 61, 182, \ldots$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #18 | Numerical Type | Award: 2 | Penalty: 0 | Discrete Mathematics

The number of positive integers less than 500 which are divisible by exactly one of 5 or 8 is ___.

Your Answer: | Correct Answer: 137:137 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #19 | Numerical Type | Award: 2 | Penalty: 0 | Discrete Mathematics

Let $S_1 = \{1, 2, 3\}$ and $S_2 = \{a, b, c, d\}$. The number of one-to-one functions from $S_1 \rightarrow S_2$ is ___.

Your Answer: | Correct Answer: 24:24 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #20 | Numerical Type | Award: 2 | Penalty: 0 | Discrete Mathematics

Number of positive odd integers less than 1000 having distinct digits is ____

Your Answer: | Correct Answer: 365:365 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #21 | Numerical Type | Award: 2 | Penalty: 0 | Discrete Mathematics

The number of non-empty partitions of a set of size 4 is ___.

Your Answer: | Correct Answer: 15:15 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #22 | Numerical Type | Award: 2 | Penalty: 0 | Discrete Mathematics

The number of ways in which we can choose 4 numbers from the set {1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13} so that more odd numbers are chosen than even is _____

Your Answer: | Correct Answer: 245:245 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #23 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Discrete Mathematics

Which of the following is the generating function corresponding to the sequence $0, 3, 1, 6, 2, 9, 3, 12, \ldots$?

A. $\frac{x(1+3x)}{(1-x^2)^2}$
B. $\frac{x(3+x)}{(1-x^2)^2}$
C. $\frac{x(2+3x)}{(1-x^2)^2}$
D. $\frac{x^2(3+x)}{(1-x^2)^2}$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #24 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Discrete Mathematics

The generating function corresponding to the sequence with closed formula for its n-th term as $a_n = 2(3^n) + 5(-1)^n$ is ___.

A. $\frac{2}{1-2x} + \frac{5}{1+2x}$
B. $\frac{2}{1-3x} + \frac{5}{1+x}$
C. $\frac{3}{1-3x} + \frac{5}{1-2x}$
D. $\frac{2}{1-4x} + \frac{5}{1-3x}$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #25 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Discrete Mathematics

The closed form solution for the n-th term ($n > 1$) of the sequence with generating function $\frac{5x}{1-4x} + \frac{2x}{1-x}$ is given by:

A. $4^n + 2$
B. $5 \cdot 4^{n-1} + 2$
C. $5 \cdot 3^{n-1} + 2$

D. none of these

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #26 | Numerical Type | Award: 2 | Penalty: 0 | Mathematical Logic

In a library, 12 identical books are to be arranged on 5 different shelves such that each shelf gets at least one book. The total number of ways in which this can be done is ___.

Your Answer: | Correct Answer: 330:330 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #27 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Discrete Mathematics

What is the middle term in the expansion of

$$\left(\frac{w\sqrt{b}}{4} - \frac{3}{b\sqrt{w}}\right)^8$$

A. $\binom{8}{4}\left(\frac{w}{b}\right)^4$

B. $\binom{8}{4}\left(\frac{b}{w}\right)^4$

C. $\binom{8}{4}\left(\frac{w}{b}\right)^2$

D. None of the above

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 02sec | Discuss

Q #28 | Numerical Type | Award: 2 | Penalty: 0 | Discrete Mathematics

The n-th term of the sequence generated by the function $\frac{3x}{(1-x)^3} \cdot \frac{2x}{1-x^2}$ is represented as a_n. The value of a_6 is _____.

Your Answer: | Correct Answer: 132:132 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #29 | Numerical Type | Award: 2 | Penalty: 0 | Discrete Mathematics

A Decimal number is called decreasing if each digit is smaller than the previous one (example, 97531). The total number of 5-digit decreasing numbers is ____.

Your Answer: | Correct Answer: 252 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #30 | Numerical Type | Award: 2 | Penalty: 0 | Discrete Mathematics

Number of rectangles that can be formed with vertices on a 4×4 grid of points is ____.

Your Answer: | Correct Answer: 100:100 | Not Attempted | Time taken: 00min 00sec | Discuss

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